Database Management

(ENCS205)

K.R. MANGALAM UNIVERSITY

Bachelor of Technology

In

Computer Science and Engineering

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SESSION: 2025-26

ASSIGNMENT 2

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

```sql
CREATE DATABASE LAB2;
USE LAB2;
DROP DATABASE LAB2;
CREATE TABLE CLASS (
    id INT PRIMARY KEY,
    name VARCHAR(50)
);
INSERT INTO CLASS VALUES (1, 'Bravo');
INSERT INTO CLASS VALUES (2, 'Alex');
INSERT INTO CLASS VALUES (4, 'Cheng');
```

2. How do you create a table named EMPLOYEES with the specified attributes?

```sql
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(100),
    Phone_Number VARCHAR(20),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary DECIMAL(10,2),
    Commission_Pct DECIMAL(5,2),
    Manager_Id INT,
    Department_Id INT
);
```

3. How do you create a table named Client_master with the specified attributes?

```sql
CREATE TABLE Client_master (
    ClientNO VARCHAR(10) PRIMARY KEY,
    Name VARCHAR(50),
    Address1 VARCHAR(100),
    City VARCHAR(50),
    Pincode INT,
    State VARCHAR(50),
    Bal_due DECIMAL(10,2)
);
```

4. What SQL statement would you use to rename the table Client_master to Client12?

```sql
ALTER TABLE Client_master
RENAME TO Client12;
```

5. How do you display the bal_due column heading as “BALANCE”?

```sql
SELECT Bal_due AS "BALANCE"
FROM Client12;
```

BALANCE

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

- INSERT INTO class VALUES (5,’Rahul’);
- COMMIT;
- UPDATE class SET name = ‘Abhijeet’ WHERE id=’5’;
- SAVEPOINT A;
- INSERT INTO class VALUES (6,’Chris’);
- SAVEPOINT B;
- INSERT INTO class VALUES (7,’Bravo’);
- SAVEPOINT C;
- SELECT * FROM class;

- ROLLBACK TO B;
- SELECT * FROM class;
- ROLLBACK TO A;

```
INSERT INTO CLASS VALUES (5,'Rahul');
COMMIT;
UPDATE CLASS SET name='Abhijeet' WHERE id=5;
SAVEPOINT A;
INSERT INTO CLASS VALUES (6,'Chris');
SAVEPOINT B;
INSERT INTO CLASS VALUES (7,'Bravo');
SAVEPOINT C;
SELECT * FROM CLASS;
```

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

```
ROLLBACK TO B;
SELECT * FROM CLASS;
ROLLBACK TO A;
SELECT * FROM class;
```

id	name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.

```
INSERT INTO EMPLOYEES VALUES
(101, 'Priya', 'bisht', 'priya@gmail.com', '1234567890', '2003-06-17', 'AD_PRES', 24000, NULL, NULL, 90),
(102, 'Neena', 'khankriyal', 'neena@gmail.com', '1234567891', '2005-09-21', 'AD_VP', 17000, NULL, 101, 90)
(103, 'Khushi', 'jindal', 'khushi@gmail.com', '1234567892', '2001-01-13', 'AD_VP', 17000, NULL, 101, 90),
(104, 'Amit', 'kumar', 'amit@gmail.com', '1234567893', '2006-01-03', 'IT_PROG', 9000, NULL, 103, 60),
(105, 'Bruce', 'den', 'bruce@gmail.com', '1234567894', '2007-05-21', 'IT_PROG', 6000, NULL, 103, 60),
(106, 'David', 'chawla', 'David@gmail.com', '1234567895', '2005-06-25', 'IT_PROG', 4800, NULL, 103, 60),
(107, 'Neha', 'null', 'neha@gmail.com', '1234567896', '2006-02-05', 'IT_PROG', 4800, NULL, 103, 70),
(108, 'Diya', 'bisht', 'Diya', '1234567897', '2007-02-07', 'IT_PROG', 4200, NULL, 103, 80);
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

Employee_Id	First_Name	Last_Name	Salary
101	Priya	bisht	24000.00
102	Neena	khankriyal	17000.00
103	Khushi	jindal	17000.00
104	Amit	kumar	9000.00
105	Bruce	den	6000.00
106	David	chawla	4800.00
107	Neha	null	4800.00
108	Diya	bisht	4200.00
NULL	NULL	NULL	NULL

2. List out the employees who work under manager 101.

```
SELECT *
FROM EMPLOYEES
WHERE Manager_Id = 101;
```

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
102	Neena	khankriyal	neena@gmail.com	1234567891	2005-09-21	AD_VP	17000.00	NULL	101	90
103	Khushi	jindal	khushi@gmail.com	1234567892	2001-01-13	AD_VP	17000.00	NULL	101	90
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Find the names of the employees who have a salary greater than or equal to 4800.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

First_Name	Last_Name
Priya	bisht
Neena	khankriyal
Khushi	jindal
Amit	kumar
Bruce	den
David	chawla
Neha	null

4. List out the employees whose last name is 'chawla'.

```
SELECT *
FROM EMPLOYEES
WHERE Last_Name = 'chawla';
```

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
106	David	chawla	David@gmail.com	1234567895	2005-06-25	IT_PROG	4800.00	NULL	103	60
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. Find the names of the employees who work in departments 60, 70, and 80.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60, 70, 80);
```

First_Name	Last_Name
Amit	kumar
Bruce	den
David	chawla
Neha	null
Diya	bisht

6. Display the unique Manager_Id.

```
SELECT DISTINCT Manager_Id
FROM EMPLOYEES;
```

Manager_Id
NULL
101
103

Part C: Updating Records (Client_master Table)

1. Insert five records into the Client_master table.

```
INSERT INTO Client12 VALUES
('C201','Rohan','Street 1','Delhi',110001,'Delhi',6000),
('C202','Meena','Street 2','Mumbai',400001,'Maharashtra',3000),
('C203','Amit','Street 3','Chennai',600001,'Tamil Nadu',8000),
('C204','Sneha','Street 4','Delhi',110002,'Delhi',2000),
('C205','Vikas','Street 5','Mumbai',400002,'Maharashtra',7000);
```

2. Find the names of clients whose bal_due > 5000.

```
SELECT Name, Bal_due
FROM Client12
WHERE Bal_due > 5000;
```

Name	Bal_due
Rohan	6000.00
Amit	8000.00
Vikas	7000.00
Rohan	6000.00
Amit	8000.00
Vikas	7000.00

3. Change the bal_due of ClientNO "C203" to Rs. 5100.

```
UPDATE Client12
SET Bal_due = 5100
WHERE ClientNO = 'C203';
SELECT * FROM Client12
WHERE ClientNO = 'C203';
```

ClientNO	Name	Address1	City	Pincode	State	Bal_due
C203	Amit	Street 3	Chennai	600001	Tamil Nadu	5100.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL